

Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : 11.9

11.9 Representations of Functions as Power Series.

(13)



11.9 Power series representation



Form	Representation	Radius of convergence
$\frac{1}{1-u}$	$\sum_{n=0}^{\infty} u^n$	$ u < 1$
$\frac{1}{1+u}$	$\sum_{n=0}^{\infty} (-1)^n u^n$	$ u < 1$
$\tan^{-1} u$	$\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n+1}}{2n+1}$	$ u < 1$
$\ln(1+u)$	$\sum_{n=0}^{\infty} (-1)^n \frac{u^{n+1}}{n+1}$	$ u < 1$

*



يستخدم هذا الجدول فقط

about

$$x = 0$$





شكل السؤال



يعطيني الدالة $f(x)$



ويطلب



ويطلب الـ *series* المساوية لها



خطوات الحل



- 1) كتابة الدالة بأحد الأشكال المعروفة.
- 2) استنتاج u
- 3) تطبيق قانون الـ *series* المساوي لها.
- 4) التبسيط "ان وجد" وحساب نصف القطر

Find the power series representation and determine the radius of convergence for

$$\frac{1}{2+x}$$

$$\begin{aligned} & \frac{1}{2 \left(1 + \frac{x}{2} \right)} \\ &= \frac{1}{2} \frac{1}{1 + \frac{x}{2}} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{2} \right)^n \\ &= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{2^n} = \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{2^{n+1}} \end{aligned}$$

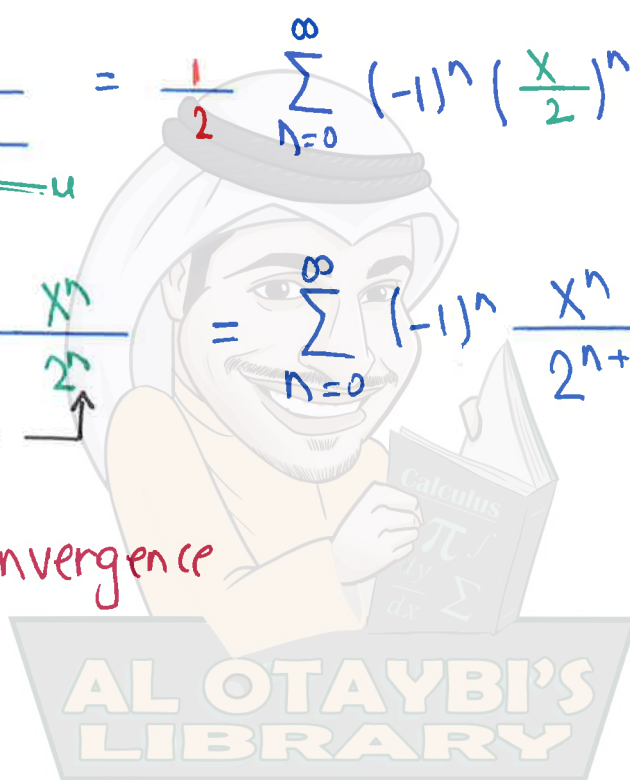
radius of convergence

$$\left| \frac{x}{2} \right| < 1$$

$$|x| < 2$$

$$-2 < x < 2$$

$$R = \frac{2 - (-2)}{2} = \frac{4}{2} = 2$$





Text book question



Find a power series representation for the function and determine the radius of convergence.

$$f(x) = \frac{x}{2x^2 + 1}$$

$$f(x) = \frac{x}{1 + 2x^2}$$

$$= x \cdot \frac{1}{1 + \underbrace{2x^2}_u}$$

$$= x \sum_{n=0}^{\infty} (-1)^n (2x^2)^n$$

$$= x \sum_{n=0}^{\infty} (-1)^n 2^n x^{2n}$$

$$= \sum_{n=0}^{\infty} (-2)^n x^{2n+1}$$

radius of convergence

$$|u| < 1$$

$$|2x^2| < 1 \rightarrow x^2 < \frac{1}{2} \rightarrow |x| < \frac{1}{\sqrt{2}}$$

$$-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}} \rightarrow R = \frac{\frac{1}{\sqrt{2}} - (-\frac{1}{\sqrt{2}})}{2} = \frac{\frac{2}{\sqrt{2}}}{2} = \frac{1}{\sqrt{2}}$$

Find the power series representation and determine the radius of convergence for

$$f(x) = \frac{x^3}{x+2}$$

$$\begin{aligned}
 x^3 \frac{1}{x+2} &= x^3 \frac{1}{2 \left[\frac{x}{2} + 1 \right]} \\
 \frac{x^3}{2} \frac{1}{1 + \frac{x}{2}} &= \frac{x^3}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{2} \right)^n \\
 \frac{x^3}{2} \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{2^n} &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+3}}{2^{n+1}}
 \end{aligned}$$

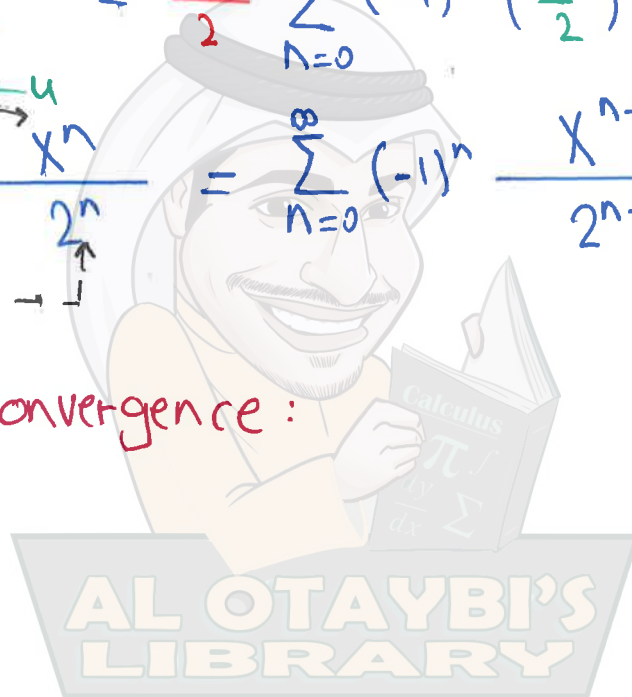
radius of convergence:

$$\left| \frac{x}{2} \right| < 1$$

$$|x| < 2$$

$$-2 < x < 2$$

$$R = \frac{2 - (-2)}{2} = \frac{4}{2} = 2$$



Find the power series representation and determine the radius of convergence for

$$f(x) = \frac{4}{3 + 2x - x^2}$$

$$- [x^2 - 2x - 3]$$

$$- [(x-1)^2 - 1 - 3]$$

$$- [(x-1)^2 - 4] = 4 - (x-1)^2$$

$$f(x) = \frac{4}{4 - (x-1)^2} = \frac{1}{1 - \frac{(x-1)^2}{4}}$$

$$= \frac{1}{1 - \frac{(x-1)^2}{4}}$$

$$= \sum_{n=0}^{\infty} \left(\frac{(x-1)^2}{4} \right)^n = \sum_{n=0}^{\infty} \frac{(x-1)^{2n}}{4^n}$$

radius of convergence:

$$\left| \frac{(x-1)^2}{4} \right| < 1 \longrightarrow (x-1)^2 < 4$$

$$|x-1| < 2$$

$$R = 2$$



Text book question



Find a power series representation for the function and determine the radius of convergence.

$$f(x) = \frac{x^3}{(x-2)^2}$$

$$\frac{1}{x-2} = \frac{1}{-2} \frac{1}{1 - \frac{x}{2}} = -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$$

$$= -\frac{1}{2} \sum_{n=0}^{\infty} \frac{x^n}{2^n}$$

$$\frac{1}{x-2} = - \sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}}$$

Diff. w.r.t. x

$$\frac{1}{(x-2)^2} = - \sum_{n=1}^{\infty} \frac{n x^{n-1}}{2^{n+1}}$$

* $(-x^3)$

$$\frac{x^3}{(x-2)^2} = x^3 \sum_{n=1}^{\infty} \frac{n x^{n-1}}{2^{n+1}}$$

$$\frac{x^3}{(x-2)^2} = \sum_{n=1}^{\infty} \frac{n x^{n+2}}{2^{n+1}}$$

radius of convergence:

$$\left| \frac{x}{2} \right| < 1 \longrightarrow |x| < 2 \longrightarrow R=2$$

Use differentiation to find a power series representation for

$$f(x) = \frac{x^3}{(2+3x)^2}$$

and determine its radius of convergence.

Spring 2019

$$\frac{1}{2+3x} = \frac{1}{2} \cdot \frac{1}{1 + \frac{3x}{2}} = u$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3x}{2}\right)^n = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n 3^n \frac{x^n}{2^n}$$

$$\frac{1}{2+3x} = \sum_{n=0}^{\infty} (-3)^n \frac{x^n}{2^{n+1}}$$

Diff. w.r.t. x

$$\frac{0 - (1)(3)}{(2+3x)^2} = \sum_{n=1}^{\infty} (-3)^n \frac{n x^{n-1}}{2^{n+1}}$$

$$\frac{-\frac{x^3}{(-3)} x^3}{(2+3x)^2} = \frac{x^3}{(-3)} \sum_{n=1}^{\infty} (-3)^n \frac{n x^{n-1}}{2^{n+1}}$$

$$\frac{x^3}{(2+3x)^2} = \sum_{n=1}^{\infty} (-3)^{n-1} \frac{n x^{n+2}}{2^{n+1}}$$

radius of convergence

$$\left| \frac{3x}{2} \right| < 1 \rightarrow |x| < \frac{2}{3} \rightarrow R = \frac{2}{3}$$

Find the power series representation of

$$f(x) = \frac{x^5}{(1+x^4)^2}, \quad |x| < 1 \text{ at } a = 0$$

$$\frac{1}{1+x^4} = \sum_{n=0}^{\infty} (-1)^n \cdot \underbrace{(x^4)^n}_{x^{4n}}$$

Diff both sides w.r.t. x

$$\frac{0 - 4x^3}{(1+x^4)^2} = \sum_{n=1}^{\infty} (-1)^n \cdot 4n x^{4n-1}$$

* x^2

$$\frac{-4x^5}{(1+x^4)^2} = \sum_{n=1}^{\infty} (-1)^n \cdot 4n x^{4n-1+2}$$

$\div (-4)$

$$\frac{x^5}{(1+x^4)^2} = \sum_{n=1}^{\infty} (-1)^n \cdot \underbrace{(-n)}_{(-1)^n} x^{4n+1}$$

$$\frac{x^5}{(1+x^4)^2} = \sum_{n=1}^{\infty} (-1)^{n+1} \cdot n x^{4n+1}$$

$$|x^4| < 1$$

$$|x| < 1$$

$$R = 1$$

Find the power series representation and determine the radius of convergence for

$$f(x) = \left(\frac{x}{2-x} \right)^3$$

$$f(x) = \frac{x^3}{(2-x)^3}$$

$$\frac{1}{2-x} = \frac{1}{2} \frac{1}{1 - \frac{x}{2}} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x}{2} \right)^n$$

$$\frac{1}{2-x} =$$

$$\sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}}$$

Diff. w.r.t. x

$$\frac{0 - (1)(-1)}{(2-x)^2} =$$

$$\sum_{n=1}^{\infty} \frac{n x^{n-1}}{2^{n+1}}$$

$$\frac{1}{(2-x)^2} =$$

$$\sum_{n=1}^{\infty} \frac{n x^{n-1}}{2^{n+1}}$$

Diff. w.r.t. x

$$\frac{0 - 2(2-x)(-1)}{(2-x)^4} =$$

$$\sum_{n=2}^{\infty} \frac{n(n-1) x^{n-2}}{2^{n+1}}$$

$$\frac{2}{(2-x)^3} = \sum_{n=2}^{\infty} \frac{n(n-1) x^{n-2}}{2^{n+1}}$$

$$* \frac{x^3}{2}$$

$$\frac{x^3}{(2-x)^3} = \frac{x^3}{2} \sum_{n=2}^{\infty} \frac{n(n-1) x^{n-2}}{2^{n+1}}$$

$$= \sum_{n=2}^{\infty} \frac{n(n-1) x^{n+1}}{2^{n+2}}$$

radius of convergence: $\left| \frac{x}{2} \right| < 1 \rightarrow |x| < 2 \rightarrow R = 2$

Find the power series representation and determine the radius of convergence for

$$f(x) = \int_0^x \frac{1}{8 - t^3} dt$$

$$\begin{aligned}
 f(x) &= \int_0^x \frac{1}{8 - t^3} dt \\
 &= \int_0^x \frac{1}{8} \frac{1}{1 - \frac{t^3}{8}} dt \\
 &= \int_0^x \frac{1}{8} \sum_{n=0}^{\infty} \left(\frac{t^3}{8} \right)^n dt \\
 &= \int_0^x \sum_{n=0}^{\infty} \frac{t^{3n}}{8^{n+1}} dt \\
 &= \sum_{n=0}^{\infty} \frac{\left[\frac{t^{3n+1}}{3n+1} \right]_0^x}{8^{n+1}} \\
 &= \sum_{n=0}^{\infty} \frac{x^{3n+1}}{(3n+1) 8^{n+1}}
 \end{aligned}$$

radius of convergence

$$\left| \frac{t^3}{8} \right| < 1 \longrightarrow |t^3| < 8$$

$$|t| < 2 \longrightarrow R = 2$$

Find the power series representation of $\frac{1}{1+x}$

Use this to find the power series representation of the function $f(x) = \tan^{-1}x$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$$

$$|x| < 1 \rightarrow R=1$$

$$\tan^{-1}x = \int \frac{1}{1+x^2} dx$$

$$= \int \sum_{n=0}^{\infty} (-1)^n (x^2)^n dx$$

$$= \int \sum_{n=0}^{\infty} (-1)^n x^{2n} dx$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$$

$$\therefore \tan^{-1}x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$$

at $x=0$

$$\tan^{-1}0 = 0 + C \longrightarrow C=0$$

$$\therefore \tan^{-1}x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

radius of convergence

$$|x^2| < 1 \longrightarrow |x| < 1 \longrightarrow R=1$$

(a) Find the power series representation of $\frac{1}{1+x}$

(b) Use this to find the power series representation of the function $f(x) = \ln(1+x^3)$

$$(a) \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n \quad |x| < 1 \rightarrow R=1$$

$$\begin{aligned} (b) \ln(1+x^3) &= \int \frac{3x^2}{1+x^3} dx \\ &= \int 3x^2 \frac{1}{1+x^3} dx \\ &= \int 3x^2 \sum_{n=0}^{\infty} (-1)^n (x^3)^n dx \\ &= \int 3 \sum_{n=0}^{\infty} (-1)^n x^{3n+2} dx \\ &= 3 \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+3}}{3n+3} + C \\ &= 3 \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+3}}{3(n+1)} + C \end{aligned}$$

$$\therefore \ln(1+x^3) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+3}}{n+1} + C$$

at $x=0$

$$\ln(1) = 0 + C \rightarrow C=0$$

$$\therefore \ln(1+x^3) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+3}}{n+1}$$

radius of convergence: $|x^3| < 1 \rightarrow |x| < 1 \rightarrow R=1$

(a) Find the power series representation of $\frac{1}{1+x^2}$

(b) Use this to find the power series representation of the function $f(x) = \int \tan^{-1} x^4 dx$

$$(a) \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n (x^2)^n \quad |x| < 1 \rightarrow R=1$$

$$\begin{aligned} (b) \tan^{-1} x^4 &= \int \frac{4x^3}{1+(x^4)^2} dx \\ &= \int 4x^3 \frac{1}{1+x^8} dx \\ &= \int 4x^3 \sum_{n=0}^{\infty} (-1)^n (x^8)^n dx \\ &= \int 4 \sum_{n=0}^{\infty} (-1)^n x^{8n+3} dx \\ &= 4 \sum_{n=0}^{\infty} (-1)^n \frac{x^{8n+4}}{8n+4} + C \end{aligned}$$

$$\therefore \tan^{-1} x^4 = 4 \sum_{n=0}^{\infty} (-1)^n \frac{x^{8n+4}}{4(2n+1)} + C$$

at $x=0$

$$0 = 0 + C \rightarrow C = 0$$

$$\int \tan^{-1} x^4 dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{8n+4}}{2n+1}$$

$$\int \tan^{-1} x^4 dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{8n+5}}{(8n+5)(2n+1)} + C$$

at $x=0$

$$0 = 0 + C \rightarrow C = 0$$

$$\begin{aligned} |x| &< 1 \\ |x| &< 1 \\ R &= 1 \end{aligned}$$

Find the power series representation of the function

$$f(x) = x^2 \tan^{-1} x$$

Apr 2016

$$\begin{aligned} f(x) &= x^2 \tan^{-1} x \\ &= x^2 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+3}}{2n+1} \end{aligned}$$

radius of convergence

$$|x| < 1 \rightarrow -1 < x < 1 \rightarrow R = \frac{1 - (-1)}{2} = 1$$

Solution #2

$$\begin{aligned} \tan^{-1} x &= \int \frac{1}{1+x^2} dx \\ &= \int \sum_{n=0}^{\infty} (-1)^n (x^2)^n dx \\ \tan^{-1} x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C \end{aligned}$$

at $x=0$

$$0 = 0 + C \rightarrow C = 0$$

$$\therefore \tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

$$\begin{aligned} \therefore f(x) &= x^2 \tan^{-1} x \\ &= x^2 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+3}}{2n+1} \end{aligned}$$

radius of convergence

$$|x^2| < 1 \rightarrow |x| < 1 \rightarrow -1 < x < 1 \rightarrow R = \frac{1 - (-1)}{2} = 1$$



Text book question



Find a power series representation for the function and determine the radius of convergence.

$$f(x) = x^2 \tan^{-1}(x^3)$$

$$\begin{aligned} f(x) &= x^2 \tan^{-1}(x^3) \\ &= x^2 \sum_{n=0}^{\infty} (-1)^n \frac{(x^3)^{2n+1}}{2n+1} \\ &= x^2 \sum_{n=0}^{\infty} (-1)^n \frac{x^{6n+3}}{2n+1} \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{6n+5}}{2n+1} \end{aligned}$$

radius of convergence

$$|x^3| < 1 \longrightarrow |x| < 1 \longrightarrow -1 < x < 1$$

$$R = \frac{1 - (-1)}{2} = 1$$

Find the power series representation and determine the radius of convergence for

$$f(x) = \frac{x - \tan^{-1} x}{x^3}$$

$$f(x) = \frac{1}{x^3} [x - \tan^{-1} x]$$

$$= \frac{1}{x^3} \left[x - \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \right]$$

أولاً $n=0$ للتأكد من أن x المحرقة.

$$= \frac{1}{x^3} \left[x - \left[x + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \right] \right]$$

$$= \frac{1}{x^3} \left[\cancel{x} - \cancel{x} - \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \right]$$

$$= -\frac{1}{x^3} \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

أدفع $-\frac{1}{x^3}$ للتأكد من العزب

$$= \sum_{n=1}^{\infty} (-1)^n \left(-\frac{1}{x^3} \right) \frac{x^{2n+1}}{2n+1}$$

$$= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n-2}}{2n+1}$$

Radius of convergence

$$|x| < 1$$

$$-1 < x < 1$$

$$R = \frac{1 - (-1)}{2} = \frac{2}{2} = 1$$

Find the power series representation and determine the radius of convergence for

$$f(x) = \frac{x - \tan^{-1}x}{x^3}$$

$$f(x) = \frac{1}{x^3} [x - \tan^{-1}x]$$

$$\tan^{-1}x = \int \frac{1}{1+x^2} dx$$

$$= \int \sum_{n=0}^{\infty} (-1)^n \frac{(x^2)^n}{x^{2n}} dx$$

$$\tan^{-1}x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$$

at $x=0$

$$0 = 0 + C \rightarrow C = 0$$

$$\rightarrow \tan^{-1}x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

$$\Rightarrow f(x) = \frac{1}{x^3} [x - \tan^{-1}x]$$

$$= \frac{1}{x^3} \left[x - \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \right]$$

$$\begin{matrix} n=0 \\ (-1)^0 \frac{x^{0+1}}{0+1} \end{matrix}$$

$$(1) x$$

$$= \frac{1}{x^3} \left[x - \left[x + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \right] \right]$$

$$= \frac{1}{x^3} \left[\cancel{x} - \cancel{x} - \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \right]$$

$$= -\frac{1}{x^3} \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

$$= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n-2}}{2n+1}$$

Radius of conv.

$$|x^2| < 1$$

$$|x| < 1$$

$$-1 < x < 1$$

$$R = 1 - \frac{(-1)}{2} = \frac{2}{2}$$

$$R = 1$$



Text book question



Evaluate the indefinite integral as a power series.

What is the radius of convergence ?

$$\int \frac{\tan^{-1} x}{x} dx$$

$$\int \frac{1}{x} \tan^{-1} x \, dx$$

$$\int \frac{1}{x} \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \, dx$$

$$\int \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{2n+1} \, dx$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)(2n+1)} + C$$

radius of convergence :

$$|x| < 1 \longrightarrow R = 1$$



Text book question



Find a power series representation for the function and determine the radius of convergence.

$$f(x) = \ln(5 - x)$$

$$\begin{aligned} f(x) &= \ln(5 - x) \\ &= \ln\left[5\left(1 - \frac{x}{5}\right)\right] \\ &= \ln 5 + \ln\left(1 + \frac{-x}{5}\right) \\ &= \ln 5 + \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{-x}{5}\right)^{n+1}}{n+1} \\ &= \ln 5 + \sum_{n=0}^{\infty} (-1)^n \frac{(-1)^{n+1} \left(\frac{x}{5}\right)^{n+1}}{n+1} \\ &= \ln 5 + \sum_{n=0}^{\infty} (-1)^n (-1)^{n+1} \frac{x^{n+1}}{5^{n+1}(n+1)} \\ &= \ln 5 - \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1) 5^{n+1}} \end{aligned}$$

radius of convergence

$$\begin{aligned} \left|\frac{-x}{5}\right| < 1 &\longrightarrow \left|\frac{x}{5}\right| < 1 \longrightarrow |x| < 5 \\ -5 < x < 5 &\longrightarrow R = \frac{5 - (-5)}{2} = \frac{10}{2} = 5 \end{aligned}$$

Find the power series representation and determine the radius of convergence for

$$f(x) = \frac{\ln(1+x)}{x^4}$$

$$\frac{1}{x^4} \ln(1+x) = \frac{1}{x^4} \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n-3}}{n+1}$$

radius of convergence

$$|x| < 1 \rightarrow R=1$$

Solution #2

$$\ln(1+x) = \int \frac{1}{1+x} dx$$

$$= \int \sum_{n=0}^{\infty} (-1)^n x^n dx$$

$$\therefore \ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} + C$$

at $x=0$

$$\ln(1) = 0 + C \rightarrow C=0$$

$$\therefore \ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$$

* $\frac{1}{x^4}$

$$\frac{1}{x^4} \ln(1+x) = \frac{1}{x^4} \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n-3}}{n+1}$$

radius of convergence: $|x| < 1 \rightarrow R=1$



Text book question



Evaluate the indefinite integral as a power series.

What is the radius of convergence ?

$$\int x^2 \ln(1+x) dx$$

$$\int x^2 \ln(1+x) dx = \int x^2 \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} dx$$

$$= \int \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+3}}{n+1} dx$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+4}}{(n+1)(n+4)} + C$$

at $x=0$

$$0 = 0 + C \longrightarrow C = 0$$

$$\therefore \int x^2 \ln(1+x) dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+4}}{(n+1)(n+4)}$$

radius of convergence

$$|x| < 1 \longrightarrow -1 < x < 1 \longrightarrow R = \frac{1 - (-1)}{2} = 1$$

Past exam

(a) Show that $\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} nx^n$

(b) Use the previous result to find the sum of $\sum_{n=1}^{\infty} \frac{n}{3^n}$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$|x| < 1 \rightarrow R=1$$

Diff w.r.t. x

$$\frac{0 - (-1)}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$$

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$$

* x

$$\frac{x}{(1-x)^2} = x \sum_{n=1}^{\infty} nx^{n-1}$$

$$\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} nx^n$$

$$\sum_{n=1}^{\infty} \frac{n}{3^n}$$

(b)

at $x = \frac{1}{3}$

$$\frac{\frac{1}{3}}{(1-\frac{1}{3})^2} = \sum_{n=1}^{\infty} \frac{n}{3^n}$$

$$\frac{\frac{1}{3}}{(\frac{2}{3})^2}$$

$$\frac{\frac{1}{3}}{\frac{4}{9}}$$

$$\frac{\frac{1}{3}}{\frac{4}{9}} = \sum_{n=1}^{\infty} \frac{n}{3^n}$$

$$x^n = \frac{1}{3^n}$$

$$x^n = (\frac{1}{3})^n$$

$$x = \frac{1}{3}$$

(a) Find the power series representation for

$$f_1(x) = \frac{1}{(1-x)^2} \text{ and } f_2(x) = \frac{1}{(1-x)^3}$$

(b) Use part a to show that $\sum_{n=2}^{\infty} \frac{n(n-1)}{2^{n-1}}$ converges and find its sum.

Past exam

$$(a) \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$|x| < 1 \rightarrow R=1$$

Diff. w.r.t. x

$$\frac{0 - (-1)}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}$$

$$\therefore f_1 = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}$$

Diff. w.r.t. x

$$\frac{0 - 2(1-x)(-1)}{(1-x)^4} = \sum_{n=2}^{\infty} n(n-1) x^{n-2}$$

$$\frac{2}{(1-x)^3} = \sum_{n=2}^{\infty} n(n-1) x^{n-2}$$

$$\therefore f_2 = \frac{1}{(1-x)^3} = \sum_{n=2}^{\infty} \frac{n(n-1) x^{n-2}}{2}$$

$$\frac{1}{(1-x)^3} = \sum_{n=2}^{\infty} \frac{n(n-1)x^{n-2}}{2}$$

at $x = \frac{1}{2}$

$$\frac{1}{(1-\frac{1}{2})^3} = \sum_{n=2}^{\infty} \frac{n(n-1)}{2^{n-1}}$$

$$\frac{1}{(\frac{1}{2})^3}$$

$$\frac{1}{\frac{1}{8}}$$

$$\therefore 8 = \sum_{n=2}^{\infty} \frac{n(n-1)}{2^{n-1}}$$

$$\therefore \sum_{n=2}^{\infty} \frac{n(n-1)}{2^{n-1}} \text{ is convergent}$$

and the sum = 8

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^{n-1}}$$

$$\frac{x^{n-2}}{2} = \frac{1}{2^{n-1}}$$

* 2

$$x^{n-2} = \frac{2}{2^{n-1}}$$

$$x^{n-2} = 2^{1-(n-1)}$$

$$x^{n-2} = 2^{1-n+1}$$

$$x^{n-2} = 2^{2-n}$$

$$x^{n-2} = 2^{-(n-2)}$$

$$x^{n-2} = (2^{-1})^{n-2}$$

$$x^{n-2} = \left(\frac{1}{2}\right)^{n-2}$$

$$x = \frac{1}{2}$$

(a) Find a power series representation for $f(x) = \tan^{-1}x$

(b) Use part(a) to show that

$$\frac{1}{\sqrt{3}} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^n} = \frac{\pi}{6}$$

Past exam

(a) $\tan^{-1}x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$

at $x = \frac{1}{\sqrt{3}}$

$$\tan^{-1}\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^n}$$

$$\frac{\pi}{6} = \frac{1}{\sqrt{3}} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^n}$$

$$\frac{1}{\sqrt{3}} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^n}$$

$$\frac{1}{3^{\frac{1}{2}}} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^n}$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^{n+\frac{1}{2}}}$$

$$x^{2n+1} = \frac{1}{3^{n+\frac{1}{2}}}$$

$$x^{2n+1} = \left(\frac{1}{3}\right)^{n+\frac{1}{2}}$$

$$(x^2)^{n+\frac{1}{2}} = \left(\frac{1}{3}\right)^{n+\frac{1}{2}}$$

$$\therefore x = \frac{1}{\sqrt{3}}$$

Find the sum of

$$\sqrt{3} \sum_{n=0}^{\infty} \frac{(-3)^n}{2n+1}$$

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

at $x = \sqrt{3}$

$$\tan^{-1} \sqrt{3} = \sqrt{3} \sum_{n=0}^{\infty} \frac{(-3)^n}{2n+1}$$

$$\frac{\pi}{3} = \sqrt{3} \sum_{n=0}^{\infty} \frac{(-3)^n}{2n+1}$$

$|x| < 1$
 $R=1$

$$3^{\frac{1}{2}} \sum_{n=0}^{\infty} \frac{(-3)^n}{2n+1}$$

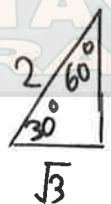
$$3^{\frac{1}{2}} \sum_{n=0}^{\infty} \frac{(-1)^n 3^n}{2n+1}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n 3^{n+\frac{1}{2}}}{2n+1}$$

$$x^{2n+1} = 3^{n+\frac{1}{2}}$$

$$(x^2)^{n+\frac{1}{2}} = 3^{n+\frac{1}{2}}$$

$$x = \sqrt{3}$$



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True or False



$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)2^{n+1}} = \ln\left(\frac{2}{3}\right)$$

$$\ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$$

$|x| < 1$
 $R=1$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)2^{n+1}}$$

$$\text{at } x = \frac{1}{2}$$

$$\ln\left(1 + \frac{1}{2}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)2^{n+1}}$$

$$\ln\left(\frac{3}{2}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)2^{n+1}}$$

$$x^{n+1} = \frac{1}{2^{n+1}}$$

$$x^{n+1} = \left(\frac{1}{2}\right)^{n+1}$$

$$x = \frac{1}{2}$$

∴ False

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